Naive Bayes Classifier.

A naive Bayes classifier is a Probabilistic machine learning model that's used for classification task.

Bayes theorem

$$\boxed{P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}} \quad \text{—— (1)}$$

{we have no. of animals pictures as a class A, given feature B is no. of features. In the test data, given feature (Black) in B.
A denote class {pictures}
B denotes feature {black} }

→ In Supervised learning model we have labelled or tagged categorize our class labels and case that a picture of elephant class labele is tagged with feature (B) Black. }. To find Black color probability in no. of classes of animals.

→ Using (1), we can find the probability of occurance of A, given B.

B → Evidence
A → Hypothesis

For eg Let's say we have Two features "headache", "cold" to predict weather person is having fever or not.

$$P(\text{fever} \mid \text{headache}, \text{cold}) = \frac{P(\text{headache} \mid \text{fever}) \times P(\text{cold} \mid \text{fever}) \times P(\text{fever})}{P(\text{headeach}) \times P(\text{cold})}$$

In above equation. the "fever" is dependent on "headache" or "cold" but these features are not dependent on each other to have "fever"

Hence the name "Naive"

The features are Independent to each other in causing fever.

Generalized formula. $X = \{x_1, x_2, x_3 \ldots x_n\}$.

$$P(y \mid x_1, x_2, \ldots x_n) = \frac{\underbrace{P(x_1|y) \times P(x_2|y) \ldots P(x_n|y)}_{\text{likelihood}} \times \underbrace{P(y)}_{\text{Prior class probability}}}{\underbrace{P(x_1) \times P(x_2) \times P(x_3) \ldots P(x_n)}_{\text{Predictor Prior Probability}}}$$

$\Downarrow$ Post erior Probability

for all entries in a data set, the denominator does not change. So it can be removed.

$$\boxed{P(y \mid x_1, x_2 \ldots x_n) \propto P(y) \prod_{i=1}^{n} (P(x_i|y))}$$

Eg Let's workout using a Small Dataset

SoNo	Type	Lengthy	Sweet	yellow	Total
1	Banana	400	350	450	500
2	Orange	0	150	300	300
3	Other	100	150	50	200
Total		500	650	800	1000

we calculate which features are more associated with which class.

(1) for Bananas

$$\frac{P(\text{Banana} \mid \text{lengthy}, \text{sweet}, \text{yellow}) \Rightarrow P(\text{lengthy} \mid \text{Banana}) \times P(\text{Sweet} \mid \text{Banana}) \times P(\text{yellow} \mid \text{Banana}) \times P(\text{Banana})}{P(\text{lengthy}) \times P(\text{sweet}) \times P(\text{yellow})}$$

= for above equation factors are →

$P(\text{lengthy} \mid \text{Banana}) = 400/500 = 0.8$

$P(\text{Sweet} \mid \text{Banana}) = 350/500 = 0.7$

$P(\text{yellow} \mid \text{Banana}) = 450/500 = 0.9$

$P(\text{Banana}) = 500/1000 = 0.5$

$P(\text{lengithy}) = 500/1000 = 0.5$

$P(\text{sweet}) = 650/1000 = 0.65$

$P(\text{yellow}) = 800/1000 = 0.8$

0.25

0.26

So substitute values in equation above

$$\cancel{P(\text{lengithy}/\text{Banana})} = \frac{0.8 \times 0.7 \times 0.9 \times 0.5}{0.5 \times 0.65 \times 0.8} = \frac{0.25}{0.26} = 0.96$$

$P(\text{Banana} / \text{lengithy}, \text{sweet}, \text{yellow}) = 0.96$

For Orange To find $P(\text{Orange} / \text{lengithy}, \text{sweet}, \text{yellow})$

$$\Rightarrow \frac{P(\text{lengithy} / \text{orange}) \times P(\text{sweet} / \text{Orange}) \times P(\text{yellow} / \text{orange}) \times P(\text{Orange})}{P(\text{lengithy}) \times P(\text{sweet}) \times P(\text{yellow})}$$

Since, $P(\text{lengithy} / \text{orange}) = 0$

$P(\text{Orange} / \text{Lengithy}, \text{sweet}, \text{yellow}) = 0$

For Other class

$P(\text{other} / \text{lengithy}, \text{sweet}, \text{yellow}) \Rightarrow \cancel{P(\text{lengithy}/\text{orange})}$

$$\frac{P(\text{lengithy} / \cancel{\text{orange}}\ \text{others}) \times P(\text{sweet} / \text{others}) \times P(\text{yellow} / \text{others}) \times P(\text{others})}{P(\text{lengithy}) \times P(\text{sweet}) \times P(\text{yellow})}.$$

$$= \frac{\frac{100}{200} \times \frac{150}{200} \times \frac{50}{200} \times \frac{200}{1000}}{0.5 \times 0.6 \times 0.8} = \frac{0.0375}{0.52}$$

$= 0.0721$

How to predict the class of fruit. (4)

We assign the class which has maximum probability which is given in eq. below

$$y = \arg\max_y P(y) \prod_{i=1}^{n} P(x_i/y) \quad \text{—} (*)$$

$P(\text{Banana} \mid \text{lengthy, sweet, yellow}) = 0.96$

$P(\text{Orange} \mid \text{lengthy, sweet, yellow}) = 0$

$P(\text{other} \mid \text{lengthy, sweet, yellow}) = 0.0721.$

The Probability of banana class is maximum (0.96) there the fruit with features lengthy, sweet, yellow is categorized in class "Banana"

~~* The class "Banana" has max probability there~~

So fruit is classified as "Banana".

So Naive Bayes classifier is algorithm based on Naive Bayes Theorem so it calculate the posterior probability which is the probability of Test data to be predicated so which assigns class. that depends on ~~includes~~ Prior feature probability & likelihood of features associated with class.

Numerical Example (2)

Naive Bayes Classifier

	Outlook	temp.	humidity	windy	Play
1	Sunny	hot	high	false	NO
2	Sunny	hot	high	T	NO
3	Overcast	hot	high	f	yes
4	Rainy	mild	high	f	yes
5	Rainy	cool	normal	f	yes
6	Rainy	cool	normal	T	NO
7	Overcast	cool	normal	T	yes
8	Sunny	mild	high	f	NO
9	Sunny	cool	normal	f	yes
10	Rainy	mild	normal	f	yes
11	Sunny	mild	normal	T	yes
12	Overcast	mild	high	T	yes
13	Overcast	hot	normal	f	yes
14	Rainy	mild	high	T	NO

yes or NO.

P(yes) = 9/14.

P(NO) = 5/14.

Individual Probabilities

(1) Outlook

	Yes	No	P(yes)	P(NO)
Sunny	2	3	2/9	3/5
Overcast	4	0	4/9	0/5
Rainy	3	2	3/9	2/5
Total	9	5	100%	100%

(2) Temperature

	Yes	No	P(yes)	P(NO)
Hot	2	2	2/9	2/5
Mild	4	2	4/9	2/5
Cool	3	1	3/9	1/5
Total	9	5	100%	100%

(3) Humidity

	Yes	No	P(Y)	P(N)
High	3	4	3/9	4/5
Normal	6	1	6/9	1/5
Total	9	5	100%	100%

(4) Wind

	Y	N	P(Y)	P(N)
False	6	2	6/9	2/5
True	3	3	3/9	3/5
	9	5	100%	100%

P / Play = yes | x) = 0.0053 | 0.0218

(5)

Play		P(yes) / P(No)
yes	9	9/14
NO	5	5/14
Total	14	100%

Probability that can play the Game.

→ P(outlook Sunny | Play = Yes) = 2/9
→ P (Temp. = cool | Play = yes) = 3/9
→ P(Humidity = High | Play = yes) = 3/9
→ P(Wind = Strong | play = Yes) = 3/9
→ P (Play = yes) = 9/14.

Probability we cannot play a Game.

→ P(Outlook = Sunny | Play = NO) = 3/5
→ P(Temperature = cool | Play = NO) = 1/5
→ P(Humidity = High | Play = NO) = 4/5
→ P(Wind. = Strong | Play = NO) = 3/5
P (Play = NO) = 5/14.

P(x|c) P(c) or P(x|Play = yes) P(Play = yes).

P(X | Play = yes) P (Play = yes) = 2/9 × 3/9 × 3/9 × 3/9 × 9/14 = 0.0053.

P(X | Play = No) P(Play = NO)
= 3/5 × 1/5 × 4/5 × 3/5 × 5/14 = 0.0206

Divide both sides to by evidence P(x) to Normalize

P(x) = P(outlook = sunny) P (Temp = cool) * P (Humid = high) * P (Wind = Strong)

P(x) = 5/14 × 4/14 × 7/14 × 6/14 = 0.02186.

Dividing the Results by this value.

$P(\text{Play} = \text{yes} \mid x) = 0.0053 / 0.02186 = 0.2424$

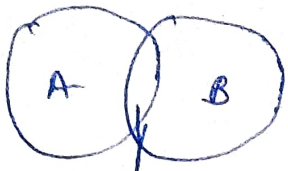
$P(\text{Play} = \text{No} \mid x) = 0.0206 / 0.02186 = 0.9421$

$\boxed{0.9421 > 0.2424}$

Ans is No to play Golf.

Bayes Theorem. — also known as Conditional Theorem

Venn Diagram



Both A & B.

How much we can Trust the evidence.

$$P(A \mid C) = \frac{P(C \mid A)\, P(A)}{P(C)}$$

Likelihood

Prior Probability

Likelihood

which describes how well the Model predicts the Data

Prior Probability

which describes the degree to which we Believe The Model ~~actually~~ accuratily Describes Reality Based on all of our Prior Inform

Posterior Probability

Which Represents the degree to which we believe a Given Model Actually describes the situation given the available data & all our prior information

Normalizing Constant

The Constant that makes the posterior Density integrate to One